

JEE Mains 2019 Chapter wise Question Bank

Gravitation - Questions

Q1

If the angular momentum of a planet of mass m , moving around the Sun in a circular orbit is L , about the center of the Sun, its areal velocity is:

- (1) $\frac{L}{m}$ (2) $\frac{4L}{m}$
 (3) $\frac{L}{2m}$ (4) $\frac{2L}{m}$

9 Jan Morning

Q2

The energy required to take a satellite to a height ' h ' above Earth surface (radius of Earth = 6.4×10^3 km) is E_1 and kinetic energy required for the satellite to be in a circular orbit at this height is E_2 . The value of h for which E_1 and E_2 are equal, is:

- (1) 1.6×10^3 km (2) 3.2×10^3 km
 (3) 6.4×10^3 km (4) 28×10^4 km

9 Jan Evening

Q3

A satellite is moving with a constant speed v in circular orbit around the earth. An object of mass ' m ' is ejected from the satellite such that it just escapes from the gravitational pull of the earth. At the time of ejection, the kinetic energy of the object is:

- (1) $2 m v^2$ (2) $m v^2$
 (3) $\frac{1}{2} m v^2$ (4) $\frac{3}{2} m v^2$

10 Jan Morning

Q4

At some location on earth the horizontal component of earth's magnetic field is 18×10^{-6} T. At this location, magnetic needle of length 0.12 m and pole strength 1.8 Am is suspended from its mid-point using a thread, it makes 45° angle with horizontal in equilibrium. To keep this needle horizontal, the vertical force that should be applied at one of its ends is:

- (1) 3.6×10^{-5} N (2) 1.8×10^{-5} N
 (3) 1.3×10^{-5} N (4) 6.5×10^{-5} N

10 Jan Evening

Q5

Two stars of masses 3×10^{31} kg each, and at distance 2×10^{11} m rotate in a plane about their common centre of mass O. A meteorite passes through O moving perpendicular to the star's rotation plane. In order to escape from the gravitational field of this double star, the minimum speed that meteorite should have at O is:

(Take Gravitational constant $G = 66 \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2}$)

- (1) 2.4×10^4 m/s (2) 1.4×10^5 m/s
 (3) 3.8×10^4 m/s (4) 2.8×10^5 m/s

10 Jan Evening

Q6

A satellite is revolving in a circular orbit at a height h from the earth surface, such that $h \ll R$ where R is the radius of the earth. Assuming that the effect of earth's atmosphere can be neglected the minimum increase in the speed required so that the satellite could escape from the gravitational field of earth is:

- (1) $\sqrt{2gR}$ (2) $\sqrt{gR}$
 (3) $\sqrt{\frac{gR}{2}}$ (4) $\sqrt{gR}(\sqrt{2}-1)$

11 Jan Morning

Gravitation

Q7

A straight rod of length L extends from $x = a$ to $x = L + a$. The gravitational force it exerts on point mass ' m ' at $x = 0$, if the mass per unit length of the rod is $A + Bx^2$, is given by:

(1) $Gm \left[A \left(\frac{1}{a+L} - \frac{1}{a} \right) - BL \right]$

(2) $Gm \left[A \left(\frac{1}{a} - \frac{1}{a+L} \right) - BL \right]$

(3) $Gm \left[A \left(\frac{1}{a+L} - \frac{1}{a} \right) + BL \right]$

(4) $Gm \left[A \left(\frac{1}{a} - \frac{1}{a+L} \right) + BL \right]$

12 Jan Morning

Q8

A satellite of mass M is in a circular orbit of radius R about the centre of the earth. A meteorite of the same mass, falling towards the earth collides with the satellite completely in elastically. The speeds of the satellite and the meteorite are the same, Just before the collision. The subsequent motion of the combined body will be

- (1) such that it escape to infinity
- (2) In an elliptical orbit
- (3) in the same circular orbit of radius R
- (4) in a circular orbit of a different radius

12 Jan Morning

Q9

Two satellites, A and B, have masses m and $2m$ respectively. A is in a circular orbit of radius R , and B is in a circular orbit of radius $2R$ around the earth. The ratio of their kinetic energies, T_A/T_B , is :

(1) $\frac{1}{2}$ (2) 1

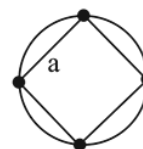
(3) 2 (4) $\sqrt{\frac{1}{2}}$

12 Jan Evening

Q10

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Four identical particles of mass M are located at the corners of a square of side ' a '. What should be their speed if each of them revolves under the influence of others' gravitational field in a circular orbit circumscribing the square ?



(1) $1.35 \sqrt{\frac{GM}{a}}$

(2) $1.16 \sqrt{\frac{GM}{a}}$

(3) $1.21 \sqrt{\frac{GM}{a}}$

(4) $1.41 \sqrt{\frac{GM}{a}}$

8 April Morning

Q11

A rocket has to be launched from earth in such a way that it never returns. If E is the minimum energy delivered by the rocket launcher, what should be the minimum energy that the launcher should have if the same rocket is to be launched from the surface of the moon? Assume that the density of the earth and the moon are equal and that the earth's volume is 64 times the volume of the moon.

(1) $\frac{E}{64}$ (2) $\frac{E}{32}$ (3) $\frac{E}{4}$ (4) $\frac{E}{16}$

8 April Evening

Q11

A solid sphere of mass ' M ' and radius ' a ' is surrounded by a uniform concentric spherical shell of thickness $2a$ and mass $2M$. The gravitational field at distance ' $3a$ ' from the centre will be:

(1) $\frac{2GM}{9a^2}$

(2) $\frac{GM}{9a^2}$

(3) $\frac{GM}{3a^2}$

(4) $\frac{2GM}{3a^2}$

9 April Morning

Q12

A test particle is moving in circular orbit in the gravitational field produced by a mass density $r(r) = \frac{K}{r^2}$. Identify the correct relation between the radius R of the particle's orbit and its period T :

(1) T/R is a constant

(2) T^2/R^3 is a constant

(3) T/R^2 is a constant

(4) TR is a constant

9 April Evening

Q13

The value of acceleration due to gravity at Earth's surface is 9.8 ms^{-2} . The altitude above its surface at which the acceleration due to gravity decreases to 4.9 ms^{-2} , is close to : (Radius of earth = $6.4 \times 10^6 \text{ m}$)

- (1) $2.6 \times 10^6 \text{ m}$ (2) $6.4 \times 10^6 \text{ m}$
(3) $9.0 \times 10^6 \text{ m}$ (4) $1.6 \times 10^6 \text{ m}$

10 April Morning

Q14

A spaceship orbits around a planet at a height of 20 km from its surface. Assuming that only gravitational field of the planet acts on the spaceship, what will be the number of complete revolutions made by the spaceship in 24 hours around the planet? [Given : Mass of Planet = $8 \times 10^{22} \text{ kg}$, Radius of planet = $2 \times 10^6 \text{ m}$, Gravitational constant $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$]

- (1) 9 (2) 17 (3) 13 (4) 11

10 April Evening

Q14

The ratio of the weights of a body on the Earth's surface to that on the surface of a planet is 9:4. The mass of the planet is $\frac{1}{9}$ th of that of the Earth. If 'R' is the radius of the Earth,

what is the radius of the planet ? (Take the planets to have the same mass density).

- (1) $\frac{R}{3}$ (2) $\frac{R}{4}$ (3) $\frac{R}{9}$ (4) $\frac{R}{2}$

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Gravitation - Answers

Q1

(3) Areal velocity $\frac{dA}{dt}$

$$dA = \frac{1}{2} r^2 d\theta$$

$$\Rightarrow \frac{dA}{dt} = \frac{1}{2} r^2 \frac{d\theta}{dt}$$

$$= \frac{1}{2} r^2 \omega$$

$$\text{Also, } L = mvr \\ = mr^2\omega$$

$$\Rightarrow \frac{dA}{dt} = \frac{1}{2} \frac{L}{m}$$

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Q2

(2) K.E. of satellite is zero at earth surface and at height h from energy conservation

$$U_{\text{surface}} + E_1 = U_h$$

$$-\frac{GM_em}{R_e} + E_1 = -\frac{GM_em}{(R_e+h)}$$

$$\Rightarrow E_1 = GM_em \left(\frac{1}{R_e} - \frac{1}{R_e+h} \right) \Rightarrow E_1 = \frac{GM_em}{(R_e+h)} \times \frac{h}{R_e}$$

Gravitational attraction

$$F_G = ma_c = \frac{mv^2}{(R_e+h)} = \frac{GM_em}{(R_e+h)^2}$$

$$mv^2 = \frac{GM_em}{(R_e+h)}$$

$$E_2 = \frac{mv^2}{2} = \frac{GM_em}{2(R_e+h)}$$

$$E_1 = E_2$$

$$\text{Clearly, } \frac{h}{R_e} = \frac{1}{2} \Rightarrow h = \frac{R_e}{2} = 3200 \text{ km}$$

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Q3

(2) At height r from center of earth, orbital velocity

$$v = \sqrt{\frac{GM}{r}}$$

By principle of energy conservation

$$\text{KE of 'm' + } \left(-\frac{GMm}{r} \right) = 0 + 0$$

(\$\because\$ At infinity, PE = KE = 0)

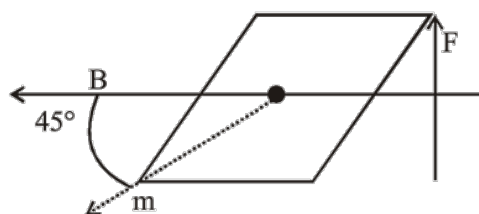
or KE of 'm'

$$= \frac{GMm}{r} = \left(\sqrt{\frac{GM}{r}} \right)^2 m = mv^2$$

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Q4

(4) using, MB sin \$\theta\$ = F \$\ell\$ Sin \$\theta\$ (\$\tau\$)



$$MB \sin 45^\circ = F \frac{\ell}{2} \sin 45^\circ$$

$$F = 2MB = 2 \times 1.8 \times 18 \times 10^{-6} = 6.5 \times 10^{-5} \text{ N}$$

10 Jan Evening

Q5

Gravitation

- (4) Let M is mass of star m is mass of meteorite
By energy conservation between 0 and ∞ .

$$-\frac{GMm}{r} + \frac{-GMm}{r} + \frac{1}{2}mV_{\text{esc}}^2 = 0 + 0$$

$$\therefore v = \sqrt{\frac{4GM}{r}} = \sqrt{\frac{4 \times 6.67 \times 10^{-11} \times 3 \times 10^{31}}{10^{11}}}$$

$$\approx 2.8 \times 10^5 \text{ m/s}$$

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Q6

- (4) For a satellite orbiting close to the earth, orbital velocity is given by

$$v_0 = \sqrt{g(R+h)} \approx \sqrt{gR}$$

Escape velocity (v_e) is

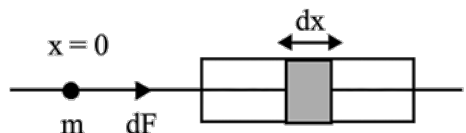
$$v_e = \sqrt{2g(R+h)} \approx \sqrt{2gR} \quad [\because h \ll R]$$

$$\Delta v = v_e - v_0 = (\sqrt{2} - 1)\sqrt{gR}$$

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Q7

- (4) Given $\lambda = (A + Bx^2)$,
Taking small element dm of length dx at a distance x from $x = 0$



$$\text{so, } dm = \lambda dx$$

$$dm = (A + Bx^2)dx$$

$$dF = \frac{Gmdm}{x^2}$$

$$\Rightarrow F = \int_a^{a+L} \frac{Gm}{x^2} (A + Bx^2) dx$$

$$= Gm \left[-\frac{A}{x} + Bx \right]_a^{a+L}$$

$$= Gm \left[A \left(\frac{1}{a} - \frac{1}{a+L} \right) + BL \right]$$

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Q8

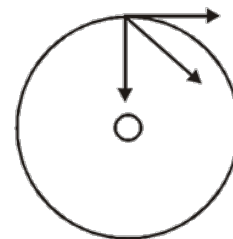
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$$(2) m\vec{v}_i + m\vec{v}_j = 2m\vec{v}$$

$$\Rightarrow \vec{v} = \frac{v}{2}\hat{i} + \frac{v}{2}\hat{j}$$

$$\Rightarrow \vec{v} = \sqrt{\left(\frac{v}{2}\right)^2 + \left(\frac{v}{2}\right)^2} = \frac{v}{\sqrt{2}}$$

$$= \frac{1}{\sqrt{2}} \times \sqrt{\frac{GM}{R}}$$



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Q9

$$(2) \text{ Orbital, velocity, } v = \sqrt{\frac{GM}{r}}$$

Kinetic energy of satellite A,

$$T_A = \frac{1}{2}m_A v_A^2$$

Kinetic energy of satellite B,

$$T_B = \frac{1}{2}m_B v_B^2$$

$$\Rightarrow \frac{T_A}{T_B} = \frac{m \times \frac{GM}{R}}{2m \times \frac{GM}{2R}} = 1$$

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Q10

Gravitation

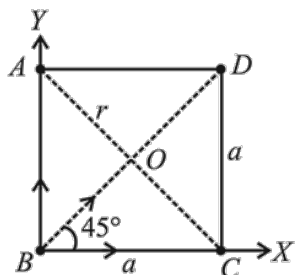
$$(2) \quad AC = a\sqrt{2} \quad \therefore r = \frac{AC}{2} = \frac{a\sqrt{2}}{2} = \frac{a}{\sqrt{2}}$$

Resultant force on the body

$$B = \frac{GM^2}{a^2} \hat{i} + \frac{GM^2}{a^2} \hat{j} + \frac{GM^2}{(a\sqrt{2})^2}$$

$$(\cos 45^\circ \hat{i} + \sin 45^\circ \hat{j})$$

$$\Rightarrow |F| = \frac{GM^2}{a^2}(\sqrt{2}) + \frac{GM^2}{2a^2}$$



$$\frac{Mv^2}{r} = \text{Resultant force towards centre}$$

$$\therefore \frac{Mv^2}{\left(\frac{a}{\sqrt{2}}\right)} = \frac{GM^2}{a^2} \left(\sqrt{2} + \frac{1}{2} \right)$$

$$\Rightarrow v^2 = \frac{GM}{a} \left(1 + \frac{1}{2\sqrt{2}} \right)$$

$$\Rightarrow v = \sqrt{\frac{GM}{a} \left(1 + \frac{1}{2\sqrt{2}} \right)} = 1.16 \sqrt{\frac{GM}{a}}$$

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Q11

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(4) Escape velocity,

$$v_c = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2G\rho V}{R}}$$

$$= \sqrt{\frac{2GS \times 4\pi R^3}{R}} = \sqrt{\frac{8}{3} \pi \rho G R^2}$$

$$\text{For moon, } v'_c = \sqrt{\frac{8}{3} \pi \rho G R_m^2}$$

$$\text{Given, } \frac{4}{3} \pi R^3 = 64 \times \frac{4}{3} \pi R_m^3 \quad \text{or } R_m = \frac{R}{4}$$

$$\therefore v'_e = \sqrt{\frac{8}{3} \pi \rho G \left(\frac{R}{4} \right)^2} = \frac{v_c}{4}$$

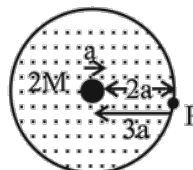
$$\frac{E}{E'} = \frac{\frac{1}{2} m v_e^2}{\frac{1}{2} m v_e'^2} = \frac{v_e^2}{v_e'^2} = \left(\frac{v_e}{v_e'} \right)^2 = 16$$

$$\text{or } E' = \frac{E}{16}$$

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Q11

$$(3) \quad E_P = \frac{GM}{(3a)^2} + \frac{G(2M)}{(3a)^2} = \frac{GM}{3a^2}$$



For a part on the surface of a spherical uniform charge distribution the whole mass acts as a point mass kept at the centre.

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Q12

Gravitation

$$\begin{aligned}
 (1) \quad F &= \frac{GMm}{r} = \int_a^R \frac{\rho(dV)m}{r^2} \\
 &= mG \int_0^R \frac{k}{r^2} \frac{4\pi r^2 dr}{r^2} \\
 &= -4\pi kGm \left(\frac{1}{r} \right)_0^R \\
 &= -\frac{4\pi kGm}{R}
 \end{aligned}$$

Using Newton's second law, we have

$$\frac{mv_0^2}{R} = \frac{4\pi kGm}{R}$$

or $v_0 = C$ (const.)

$$\text{Time period, } T = \frac{2\pi R}{v_0} = \frac{2\pi R}{C}$$

$$\text{or } \frac{T}{R} = \text{const.}$$

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Q13

(1) Given

Acceleration due to gravity at a height h from earth's surface is

$$g_h = g \left(1 + \frac{h}{R_e} \right)^{-2} \Rightarrow 4.9 = 9.8 \left(1 + \frac{h}{R_e} \right)^{-2}$$

$$\frac{1}{\sqrt{2}} = \left(1 + \frac{h}{R_e} \right) \quad [\text{as } h \ll R_e]$$

$$h = R_e (\sqrt{2} - 1)$$

$$h = 6400 \times 0.414 \text{ km}$$

$$h = 2.6 \times 10^6 \text{ m}$$

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Q14

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(4) Time period of revolution of satellite,

$$T = \frac{2\pi r}{v}$$

$$v = \sqrt{\frac{GM}{r}}$$

$$\therefore T = 2\pi r \sqrt{\frac{r}{GM}} = 2\pi \sqrt{\frac{r^3}{GM}}$$

Substituting the values, we get

$$T = 2\pi \sqrt{\frac{(202)^3 \times 10^{12}}{6.67 \times 10^{-11} \times 8 \times 10^{22}}} \text{ sec}$$

$$T = 7812.2 \text{ s}$$

$$T \approx 2.17 \text{ hr} \Rightarrow 11 \text{ revolutions.}$$

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Q14

$$(4) \quad \frac{W_e}{W_p} = \frac{mg_e}{mg_p} = \frac{9}{4}$$

$$\text{or } \frac{g_e}{g_p} = \frac{9}{4}$$

$$\text{or } \frac{GM/R^2}{G(M/9)/R_p^2} = \frac{9}{4}$$

$$\therefore R_p = \frac{R}{2}$$

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